HACK SMART @ KIT

HACKATHON

Hack Smart @ KIT, Coimbatore

(Open to all, let's innovate, together)

November 14th and 15th 2025
A 36 Hour Grand Hackathon

Guidelines

1. Open to all KIT, Coimbatore students.
2. No participation fee.
3. Each team must have 5 members.
4. Teams must be interdisciplinary – Max 2 students from the same year/department.
5. At least one female participant per team.
6. Hackathon will be conducted in-person on the KIT campus.
7. Teams can choose their own problem statements under the given themes.
8. Industry jury will evaluate and their decision will be final.
9. Teams must register online (only the team leader needs to register).
10. Your idea should be original and any sort of cheating leads to disqualification.

Rounds:

- Round 1 – Idea Submission: Submit your idea in the prescribed format.
- Round 2 – Shortlisting: Jury selects the best ideas for the next stage.
- Round 3 – Product Development & Finale – November 14th and 15th 2025.
    - Build your prototype/proof of concept with support from organizers.
    - Present and pitch your final solution to the jury.
    - Results & celebrations to follow.

Evaluation Criteria

🏆 KIT Hackathon – Prizes & Awards

Special Awards

<u>Techy awards</u>

Best 5 to 10 Tech members will be provided with International Certification voucher (full exam fee waiver) for AWS, Azure, Cybersecurity, CISCO

<u>KIT Hackathon Problem Statements on Sustainability & SDGs:</u>

1. Smart Water Management System

SDG 6 – Clean Water & Sanitation
Develop a low-cost IoT-based solution to monitor and optimize water usage in campus or residential buildings using sensors and data analytics.

2. Waste-to-Energy Converter

SDG 7 – Affordable & Clean Energy
Design a compact machine or system that converts organic or plastic waste into usable energy (biogas/electricity).

3. AI-Powered Food Waste Tracker

SDG 12 – Responsible Consumption & Production
Build a mobile or AI-based app that tracks, predicts, and minimizes food waste in hostels, canteens, or restaurants.

4. Sustainable Mobility for Campus Transport

SDG 11 – Sustainable Cities & Communities
Create an electric or hybrid vehicle design or app-based ride-pooling solution to reduce carbon footprint on campus.

5. Intelligent Energy Audit System

SDG 13 – Climate Action
Develop an AI or IoT-based energy monitoring and recommendation system for academic or industrial buildings.

6. Air Quality Monitoring and Forecasting System

SDG 3 – Good Health & Well-being
Design a distributed air pollution sensing network with predictive analytics for urban areas or industrial zones.

7. Smart Agriculture for Resource Optimization

SDG 2 – Zero Hunger
Implement a precision farming solution using sensors, drones, or ML models to optimize irrigation, fertilizer use, and crop yield.

8. Eco-friendly Construction Materials

SDG 9 – Industry, Innovation & Infrastructure
Design an innovative, low-cost, and sustainable building material using waste or by-products (fly ash, coconut fiber, etc.).

9. AI-Driven Flood or Drought Prediction System

SDG 13 – Climate Action
Use satellite data, sensors, and ML models to forecast and visualize flood-prone or drought-prone zones.

10. Plastic Recycling & Tracking Ecosystem

SDG 12 – Responsible Consumption & Production
Develop a blockchain or IoT-based system to trace plastic waste collection, sorting, and recycling lifecycle.

11. Energy-efficient Smart Street Lighting

SDG 7 – Affordable & Clean Energy
Build a sensor-based adaptive street lighting system that adjusts brightness based on motion and ambient light.

12. Campus Carbon Footprint Analyzer

SDG 13 – Climate Action
Develop a dashboard that quantifies carbon emissions from campus utilities, transport, and energy use with reduction suggestions.

13. AI-based Water Quality Prediction System

SDG 6 – Clean Water & Sanitation
Use sensor data and ML to assess water quality and detect contamination sources in real-time.

14. Inclusive Assistive Technology for Differently-abled Persons

SDG 10 – Reduced Inequalities
Design affordable devices or AI-powered systems to assist physically challenged individuals in education or workplace settings.

15. Smart Renewable Microgrid Management System

SDG 7 – Affordable & Clean Energy
Create a simulation or prototype of a microgrid integrating solar, wind, and battery systems for rural or institutional use.

Judging Criteria (out of 100)

Important Dates:

What other than prizes you get?

💡 Patent Opportunity: Outstanding ideas may be supported for patent filing through KIT's innovation cell.

🚀 Startup Funding: Promising solutions stand a chance to receive seed funding and incubation support to kickstart your entrepreneurial journey.

📢 Spotlight & Recognition: Get featured on KIT's official LinkedIn and media handles — showcase your innovation to a global audience.

Industry Mentorship: Gain direct mentoring and networking opportunities with top industry experts and innovators.

Resources:

How to win a hackathon?

How to generate ideas?

[Registration Link](https://forms.gle/u8drsn5PtnrSg4a27)

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Dept Coordinators:

Name Of The Faculty	Department
Dr.K.Anusindhiya Mr.D.Santhosh Kumar	AERO
Mr.K.Sathyaseelan Ms.R.Kavitha	AI&DS
Dr.N.Prakash Ms.M.Gokila	ECE
Mr,T.Barath Kumar Dr.K.Saranya	EEE
Dr.ArivumaniRavanan Dr.P.Hariprasad	MECH
Dr.S.Saju Ms.D.R.Nivetha	BME
Dr.B.Gopalsamy Ms.R.Subashini	BT
Dr.J.Senthil Kumar Ms.R.Thabathi	CSE
Ms.S.Savitha Mr.S.Deeban	AIML
Mr.P.Mathivanan Ms.SithikaSeema	CSBS
Mr.S.Selvin Rajesh Kumar Mr.K.Rajaram	CYBER SECURITY
Ms.S.Sheeba	VLSI